

INVESTIGATING THE USE OF IMPEDANCE PLETHYSMOGRAPHY FOR DETECTING DECREASED BLOOD FLOW IN DIABETIC PATIENT LIMBS



Mirza Taimoor Sultan BAIG

Advisor: Assoc. Prof. Mustafa İlker Beyaz

JANUARY 2023

Outline

- **Introduction**
- **Impedance Plethysmography**
- **Motivation**
- **Literature Review**
- **Contribution**
- **COMSOL Multiphysics 5.6**
- **Geometrical Model**
- **Theoretical Calculation**
- **Tissue properties**
- **Simulation Result**
- **Circuit Diagram**
- **Physical Implementation**
- **Experimental Result**
- **Conclusion**

Modern Methods of Measuring Blood Flow



Ultrasound transducer

Source: (Wearable Compact Blood Flow Sensor _ MEMS_ Microfabricated Products _ NTTAT Advanced Technology Product Introduction Site, n.d.)

- This system is made up of an app specifically designed for smartphones and a sensor gadget.
- Visualizations such as animations of heartbeats or blood flow graphs are then generated by the smartphone based on this data.



Digital Sphygmomanometer

Source: (Buy Saint Health Arm Automatic Blood Pressure Monitor BP Sphygmomanometer Pressure Meter Tonometer for Measuring Arterial Pressure at Affordable Prices — Free Shipping, Real Reviews Wi, n.d.)

- It is a digital blood pressure monitor.
- It is portable, easy to use, and rapidly and accurately detects blood pressure and pulse.



Sphygmomanometer

Source: (Aneroid Vérnyomásmérő És Sztetoszkópkészlet Egészségügyi Kézi Otthoni Vérnyomásmérő Széria Mandzsetta És Sztetoszkóp _ Wish, n.d.)

- It is also called blood pressure cuffs or blood pressure cuffs.
- Cuffs are used to stop blood flow in arteries by wrapping them around the upper arm and inflating them.

Modern Methods of Measuring Blood Flow



PPG Wristwatch

Source: Reader, R. (2022). Apple partner is developing a wearable to track blood sugar. In Fast Company. Retrieved from <https://www.fastcompany.com/90730050/apple-partner-rockley-photonics-blood-sugar>

- A Non-Intrusive method for taking blood pressure using PPG Technique.
- It can be worn on the wrist



Pulse Oximeter

Source: (SPO2 Finger Pulse Oximeter From a Reputed Online Medical Supply Store _ HealthConnection, n.d.)

- Oximeters are tiny, non-invasive gadgets that may connect to a finger or toe.
- It can tell how well the heart is supplying blood to the hands and feet.

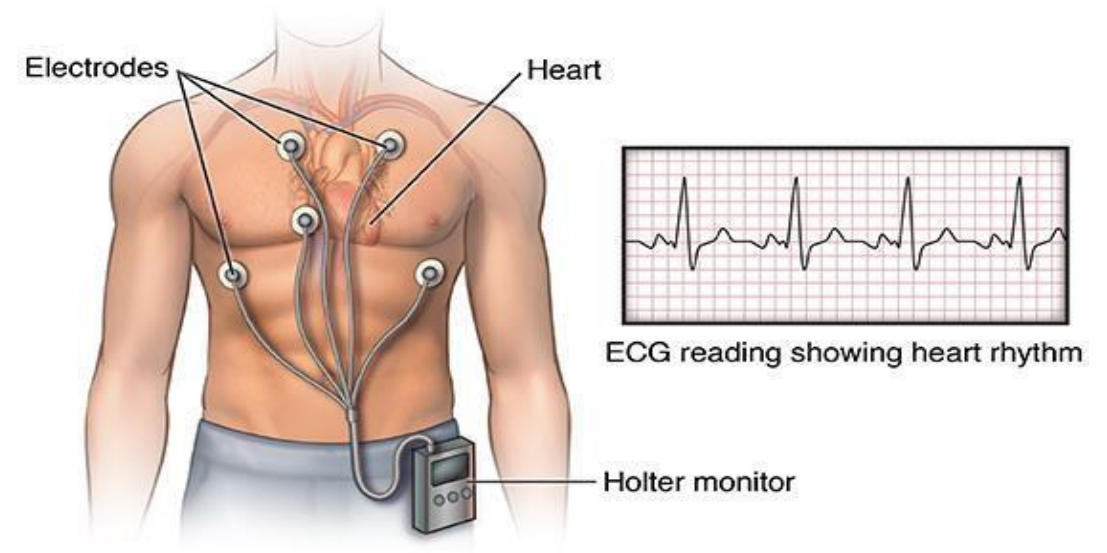
Modern Methods of Measuring Blood Flow



Electrocardiogram ECG

Source: (USA Medical and Surgical Supplies, 2018)

- At an early age of minor cardiovascular issues and signs of initial heart malfunction, an electrocardiogram or ECG is a simple, fast, and common test used to diagnose and catch an overview of the problem.
- It is a safe procedure that captures the electrical activity of the heart.



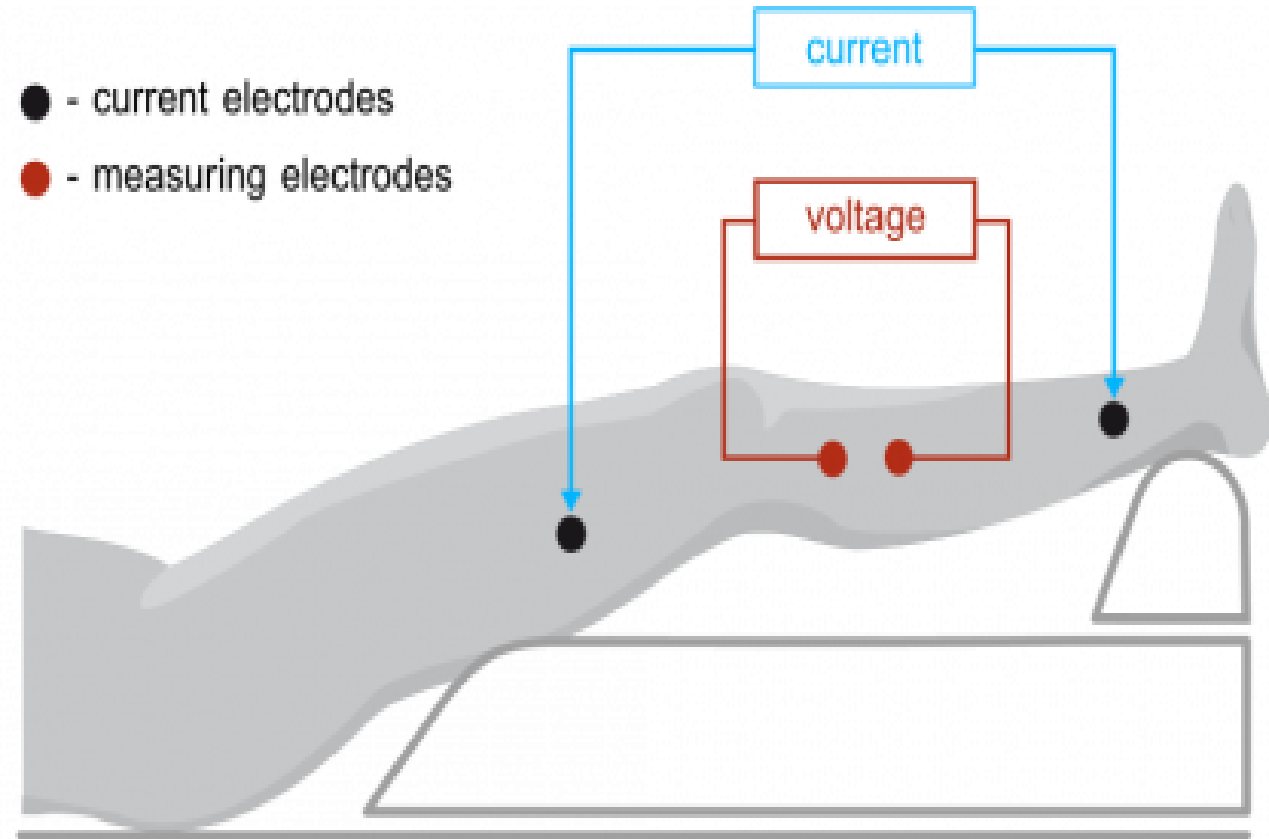
Holter Meter

Source: (Johns Hopkins Medicine, 2019)

- A Holter blood pressure device is worn by the patient for twenty-four hours.
- If the increase in blood pressure readings is to rule out or confirm the diagnosis of hypertension .

What is impedance plethysmography?

- A method that can measure plethysmography and identify the arterial pulse waveform is called bioelectrical impedance. this technique is collectively known as impedance plethysmography.
- By infusing a minute amount of AC into the body and then detecting the potential caused by blood flow, it gauges changes in blood volume.



Source: (Impedance Plethysmography (IPG) - Medis, n.d.)



Why Impedance Plethysmography?

Technique	Measure	Advantage	Disadvantage
Photoplethysmography (PPG)	● Blood Volume ● NIRS (Blood Flow estimation)	● Non-Invasive ● Easy to use ● High accuracy	● Small Volume of tissue measurement ● Localized measurement
Doppler Ultrasound	● Instant blood flow estimation ● Blood Velocity	● Non-Invasive ● Portable ● Low energy consumption	● Skilled Operation ● Single vessel measurement ● Low-depth penetration
Impedance Plethysmography (IPG)	● Blood Volume ● Blood Flow Estimation	● Non-Invasive ● Portable ● Low energy consumption ● Easy to Use ● Measure a large number of tissues	● Signal close to the noise floor

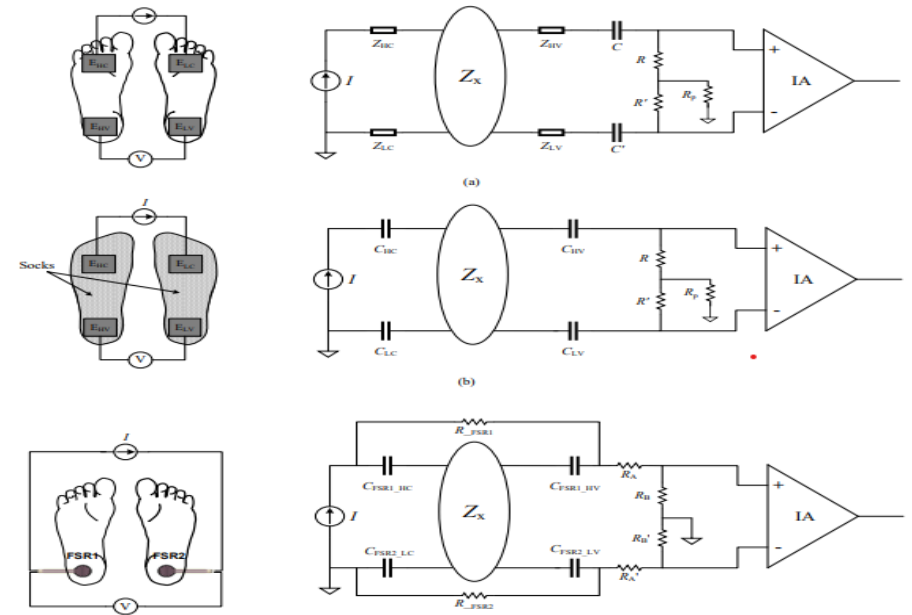
Motivation



- **Monitoring the blood flow device is not affordable for everyone.**
- **Expensive devices need to be precise and close to accuracy for the patient, however, it is directly proportional to the price they pay.**
- **Everyone especially diabetic patients can measure their existence if the blood flow at home.**
- **When the blood flow stops this device will give a signal that you should go to a doctor immediately.**
- **To make a Portable and wearable device that everyone can afford easily.**

Literature Review

- According to Isabel Morales' research, the foot impedance plethysmography was carried out with two distinct kinds of electrodes, using 1mA/10kHz and finding the impedance of the foot.
- In this study Piuzzi research is an ultimate technique selected that makes use of a current with a frequency of 50 kHz that is injected into the thorax of the subject using a couple of textile electrodes.
- In this study of Ramkumar & Babu, blood flow was performed using a fuzzy logic toolbox with many different activities, measured before and after the 25 seconds of breath-holding activity. The plethysmographic waveform of forearm impedance can be used to analyze the changes in a heartbeat that are correlated with the changes in heart stroke volume.



Source: Morales, I., Gonzalez-Landaeta, R., & Simini, F. . Pressure sensors used as bioimpedance plantar electrodes: A feasibility study. 2021 IEEE International Symposium on Medical Measurements and Applications,

Contribution

- **In this work we are investigating using multiple electrodes to see the additional data going to make testing easier.**
- **Our research is primarily focusing on the amount of blood flow through the finger of the upper limb for diabetics patients.**

COMSOL Multiphysics 5.6

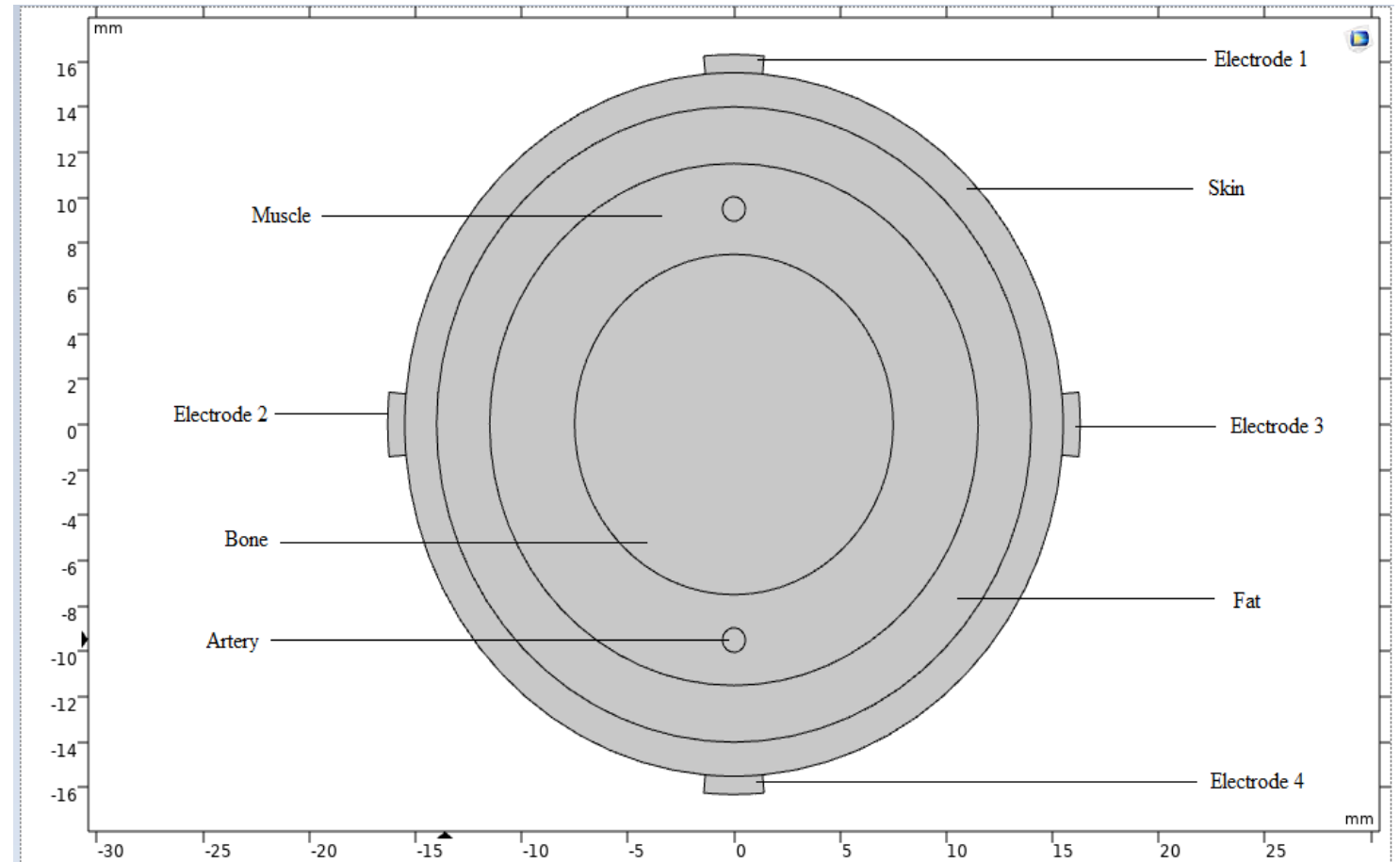
- We are using COMSOL Model physics to investigate the tests.
- The geometrical model of the test is on the 2D dimension.
- In the COMSOL we are using electrical current physics and we are choosing the frequency domain.



Geometrical Model



- This geometry model was constructed to demonstrate the internal structure of the forefinger.
- Electrodes numbering 1,2,3 and 4 are attached to the surface of the forefinger.
- This way it is easier for our software to detect the best frequency penetrated across the layers.



Frequency-Dependent Tissue Properties



- The layers presented in the geometrical model namely: skin, fat, muscle, bone, and artery will produce different values at different frequencies.
- A theoretical view of these values is shown.
- Depicting permittivity and conductivity.

The dielectric properties of tissues at different frequencies:

MATERIALS	10Hz	100Hz	1KHz	10KHz	50KHz
Skin	0.0002	0.000200001	0.000200065	0.000204082	0.000273088
Fat	0.012206696	0.020810608	0.022404607	0.023830606	0.024246193
Muscle	0.201966739	0.266709174	0.321156604	0.340831688	0.351827808
Bone	0.075563518	0.081031622	0.081529994	0.082623046	0.083421828
Artery	0.7	0.700000006	0.700000484	0.700038264	0.700800987

(a) Electrical conductivity (S/m)

MATERIALS	10Hz	100Hz	1KHz	10KHz	50KHz
Skin	1135.9904	1135.9398	1135.6193	1133.5616	1126.7574
Fat	7973504.661	457062.2399	24104.2143	1085.2981	172.4172
Muscle	25700364.51	9329046.966	434932.3153	25908.7129	10093.7268
Bone	10020467.96	217032.9911	12320.0379	1657.7721	613.1811
Artery	5259.9782	5259.8264	5258.6084	5248.1698	5197.6999

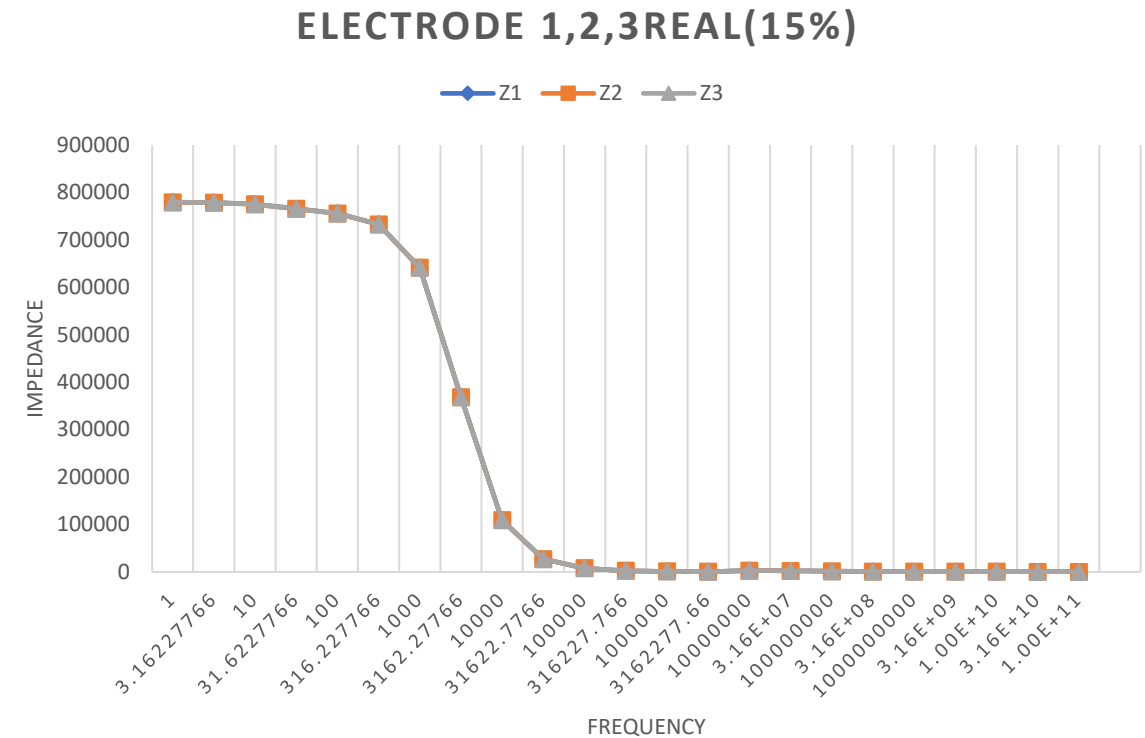
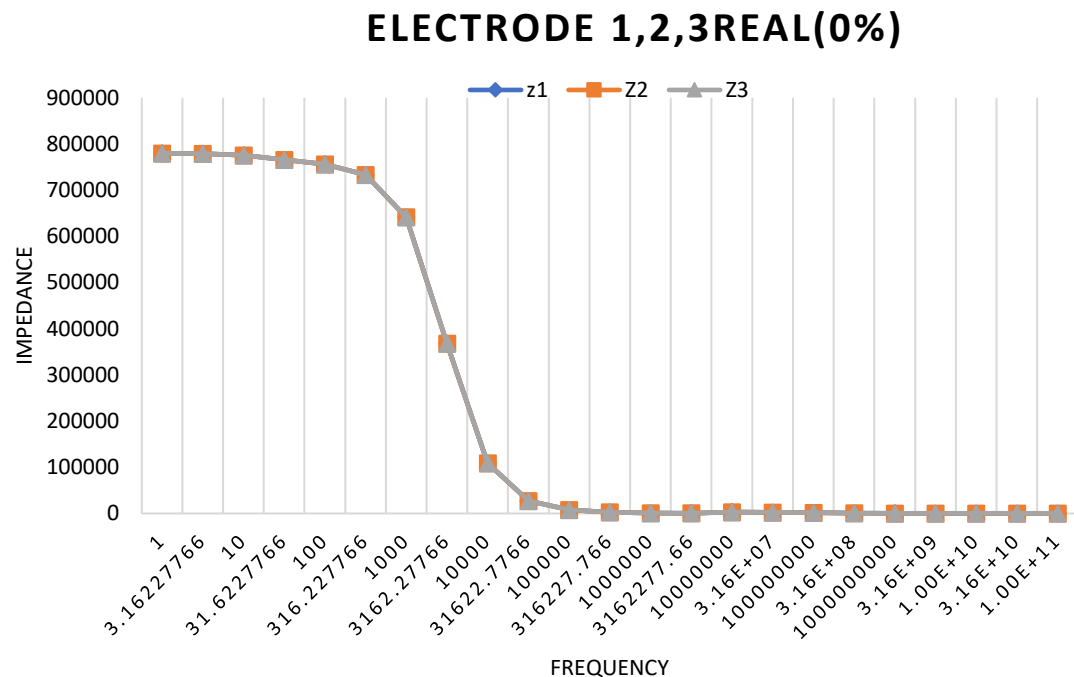
(a) Electrical permittivity

Source: (Tissue Frequency Chart » IT'IS Foundation, n.d.)

Simulation Results



- When the heart is diastolic condition.
- When the heart is systolic condition.

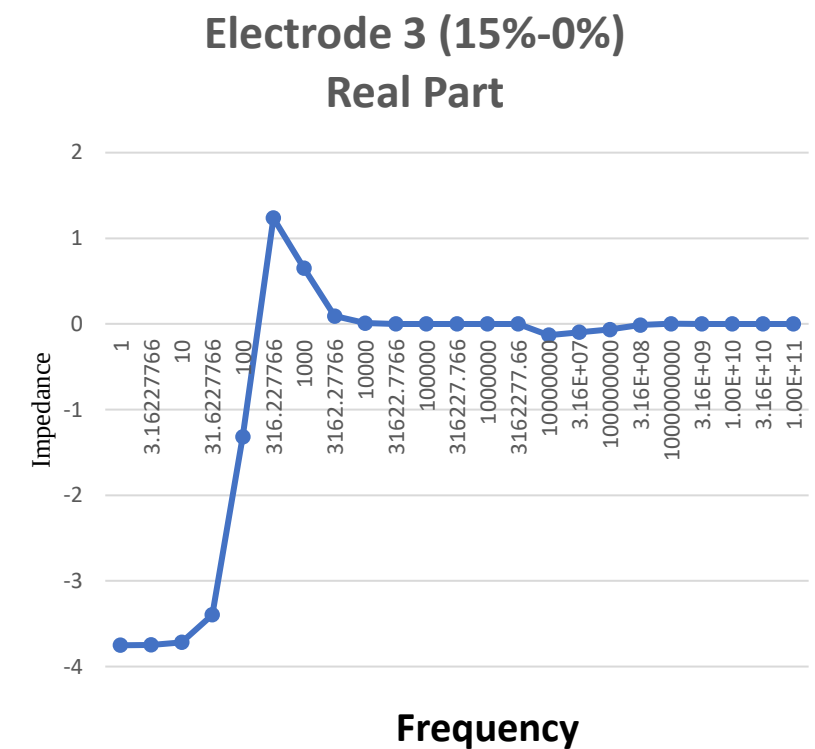
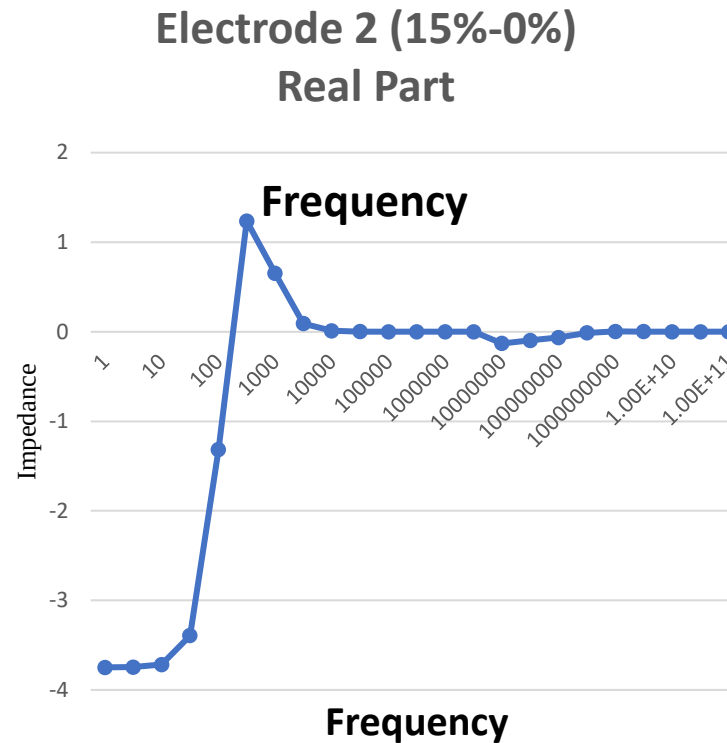
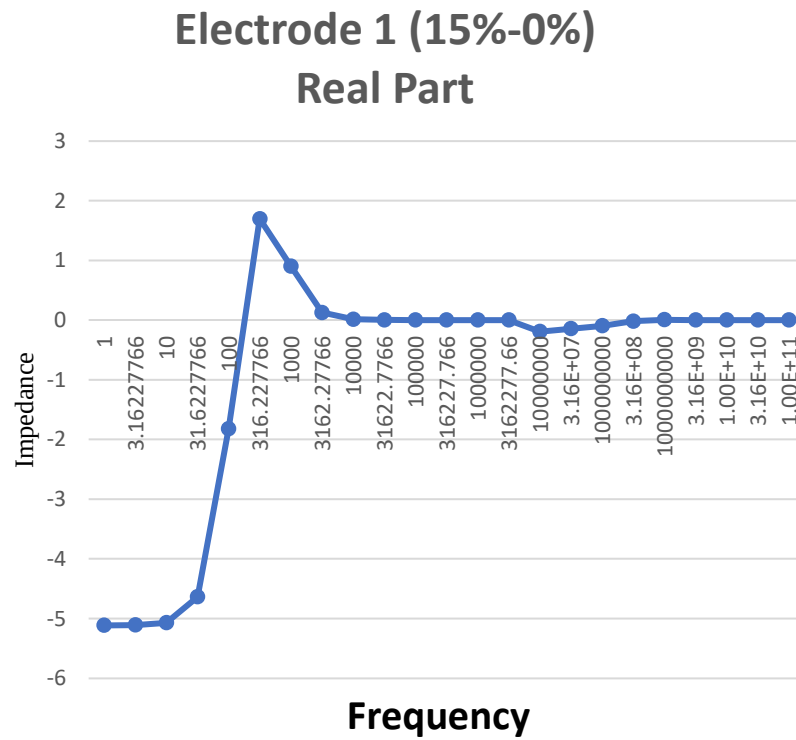


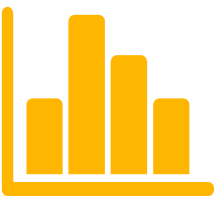
Artery expends 15% of its original value, because when the heart pump and supplies the blood to artery then the artery size will be expended.



Simulation Results

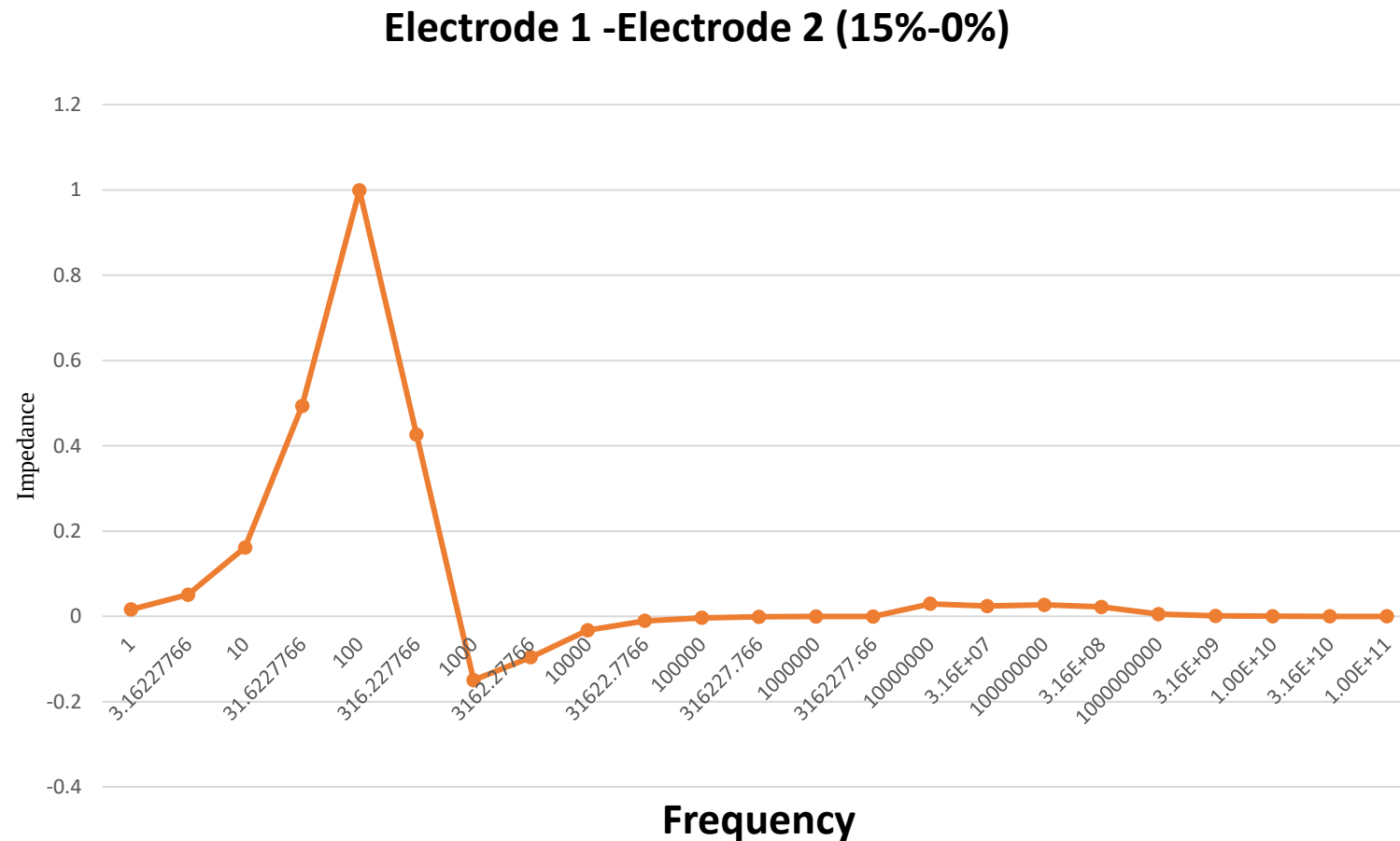
- The simulation result shows us that the output of electrodes 2 and 3 has given nearly the same values. Therefore, we would choose one out of electrodes 2 and 3.



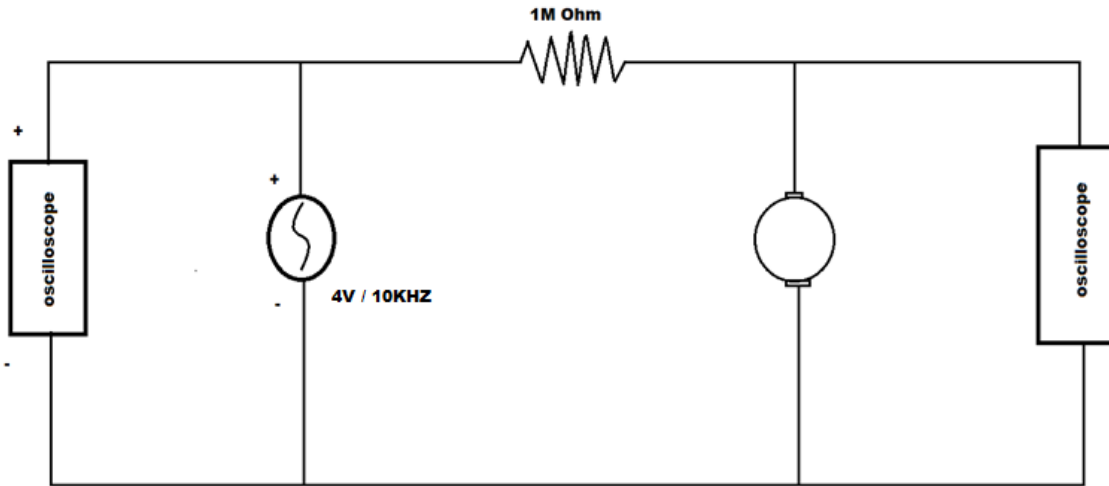


Simulation Results

- Finally, we took 100 Hz as the optimal frequency to get 1 as our result.

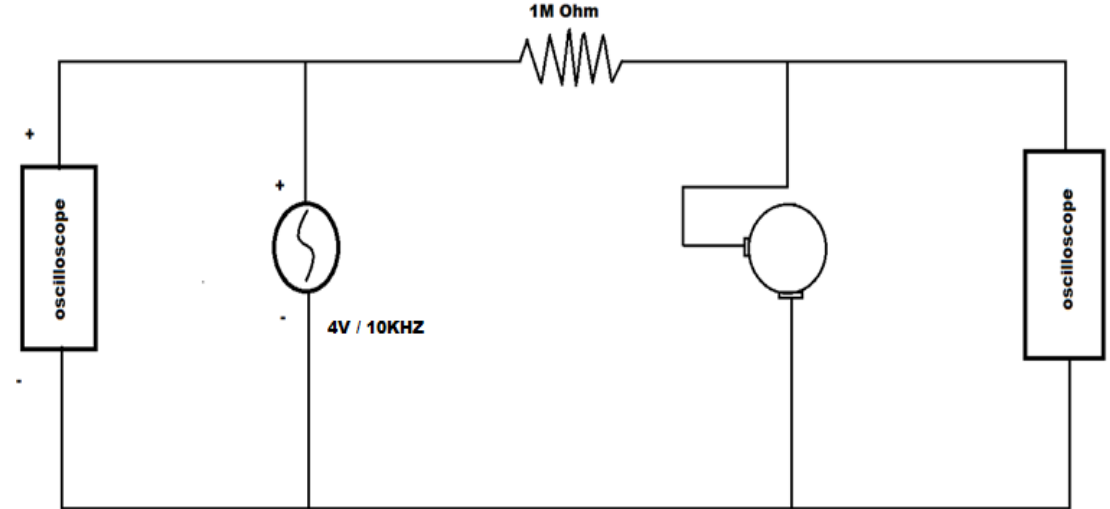


Circuit Diagram

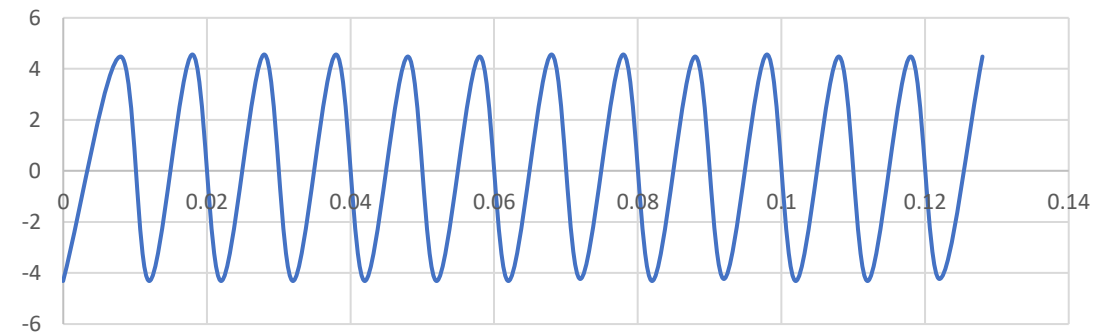


Block Diagram of Position of electrodes top-bottom

- We are using 10mA with respect to frequency 100Hz /10KHz.
- We try to look at the changes in peak-to-peak voltages.

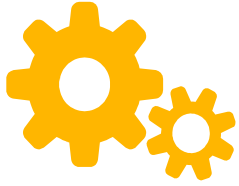


Block Diagram of Position of electrodes Side-bottom

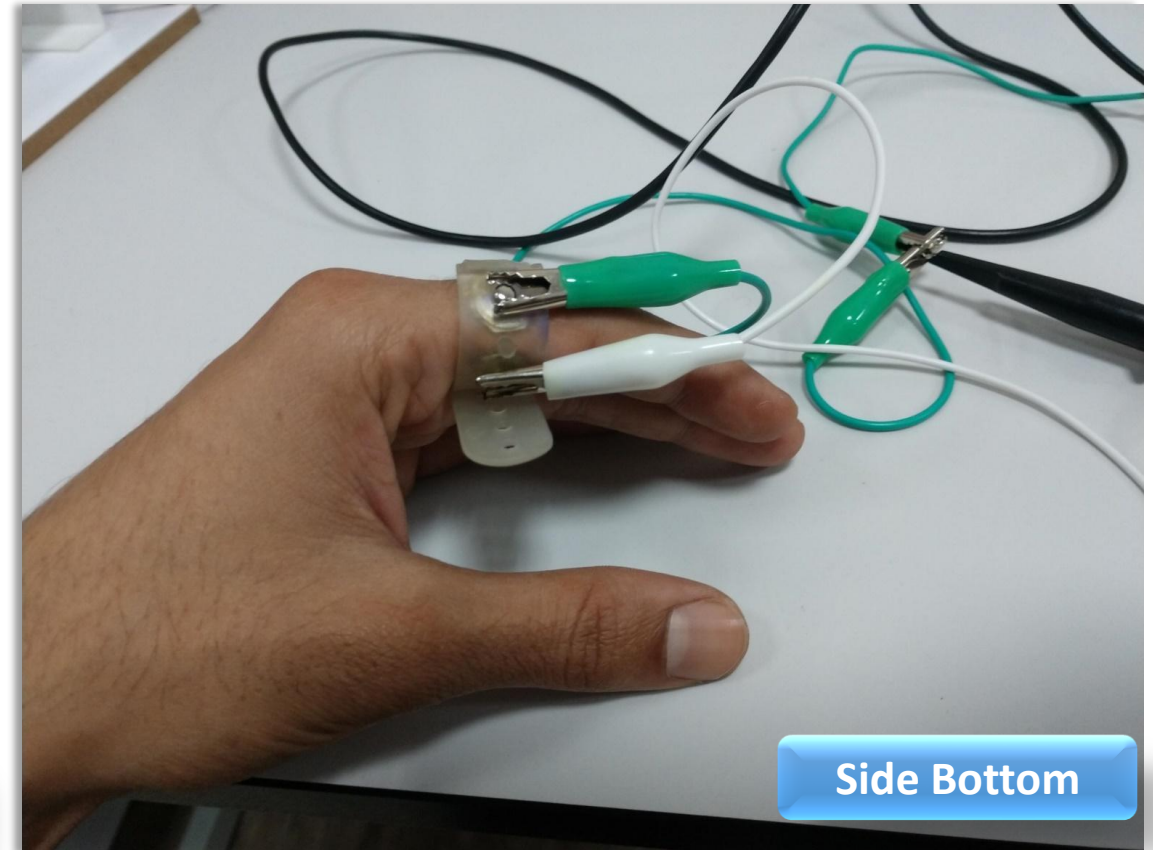
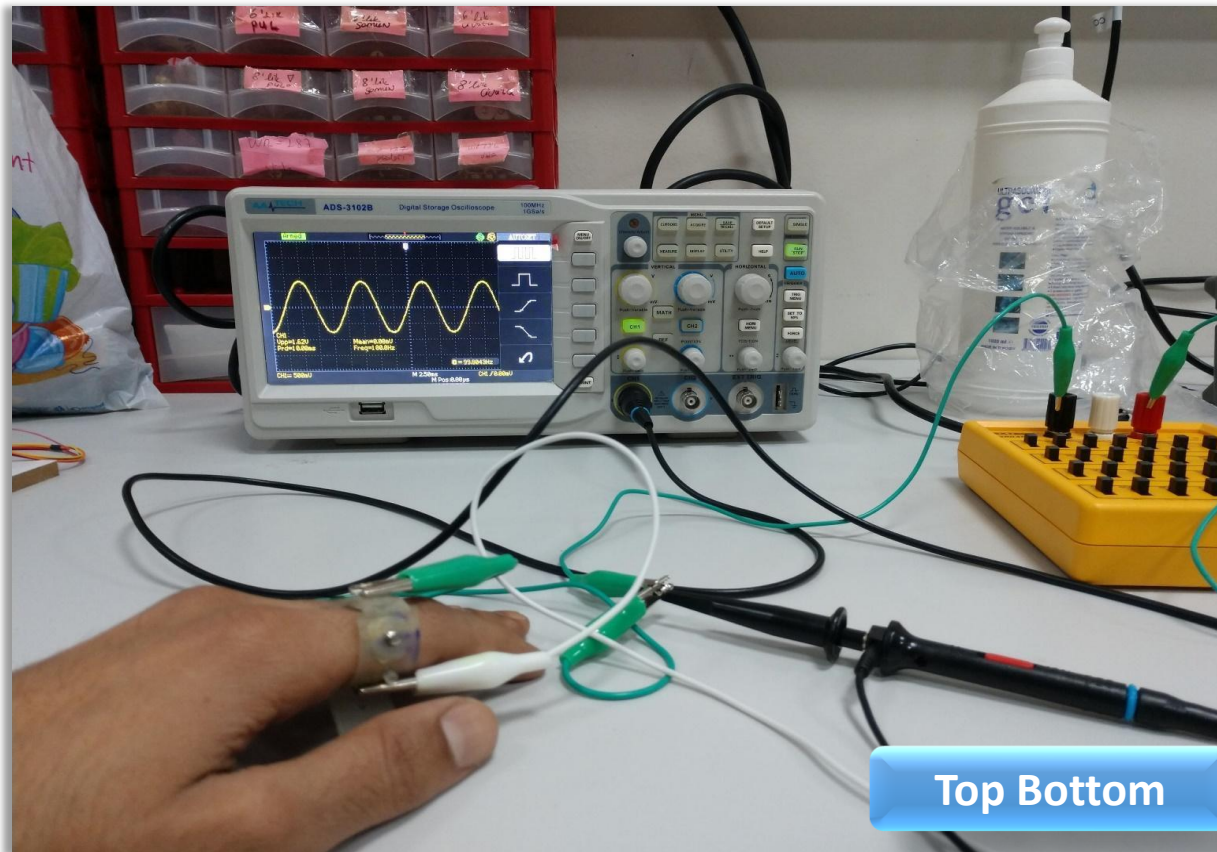


Oscilloscope

Physical Implementation



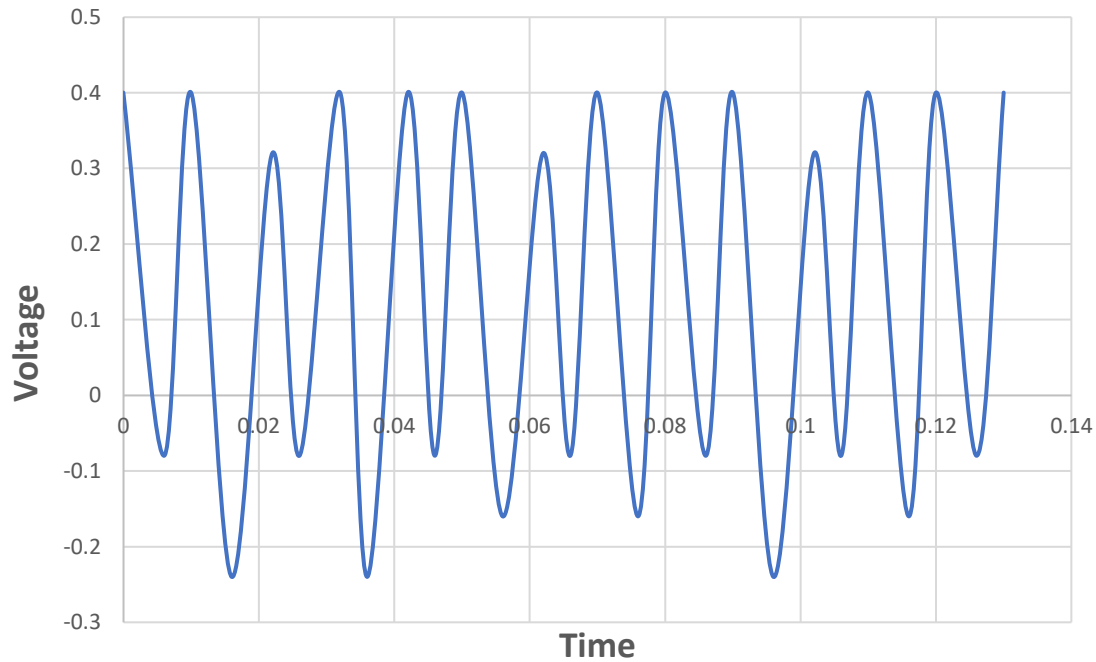
- Visualization of the electrodes clipped on the forefinger.



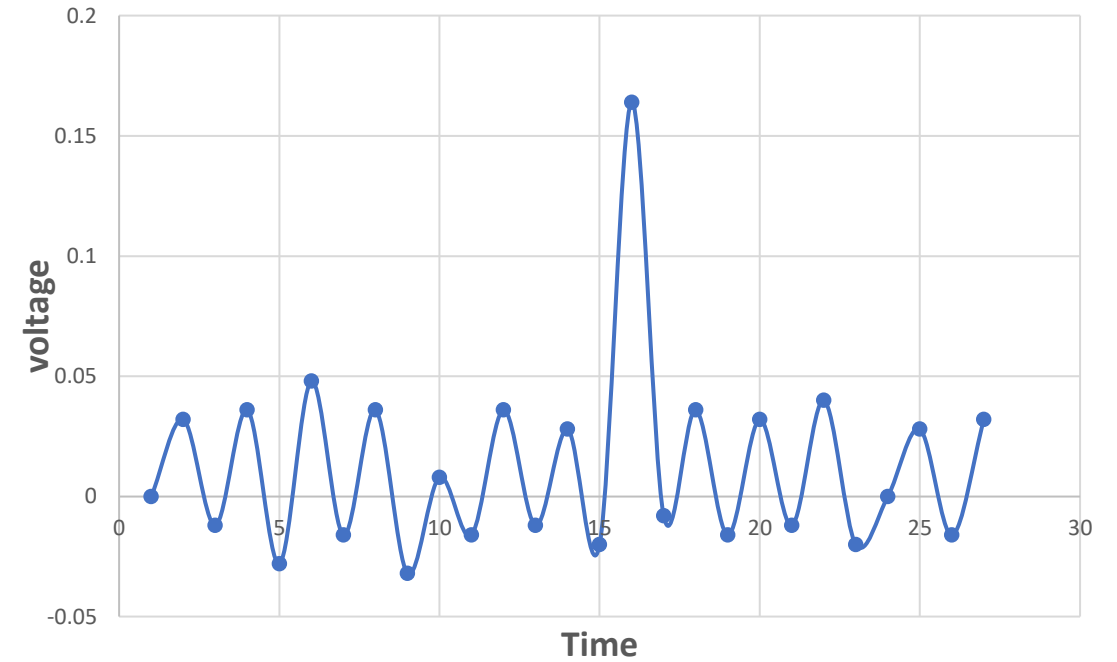
Top-Bottom Results



- Resulting graphs of the top bottom with 100Hz and 10KHz.



Using 100Hz Frequency

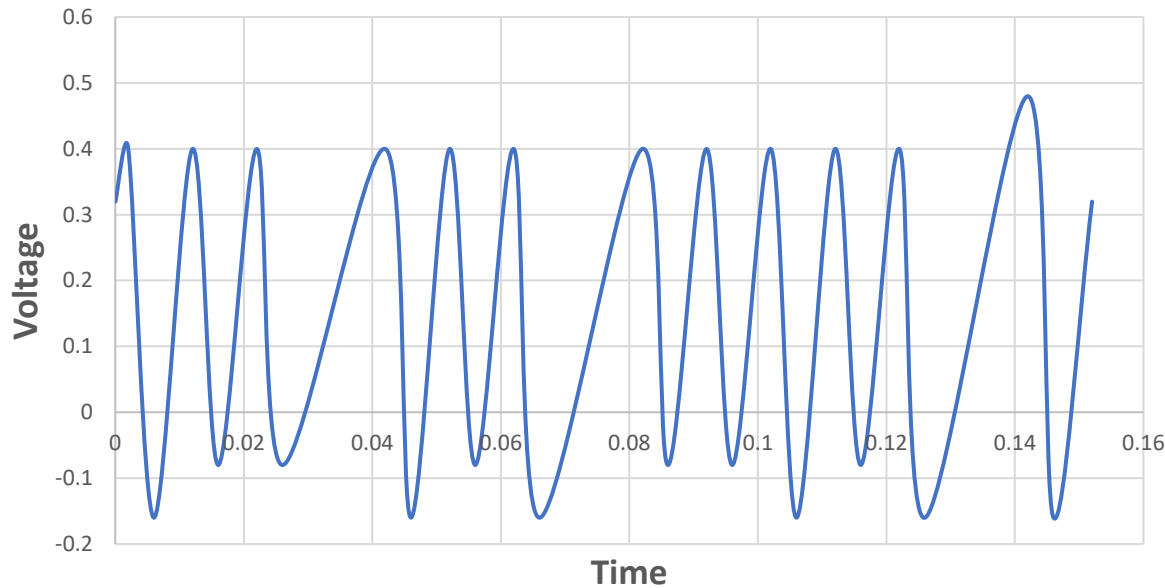


Using 10KHz Frequency

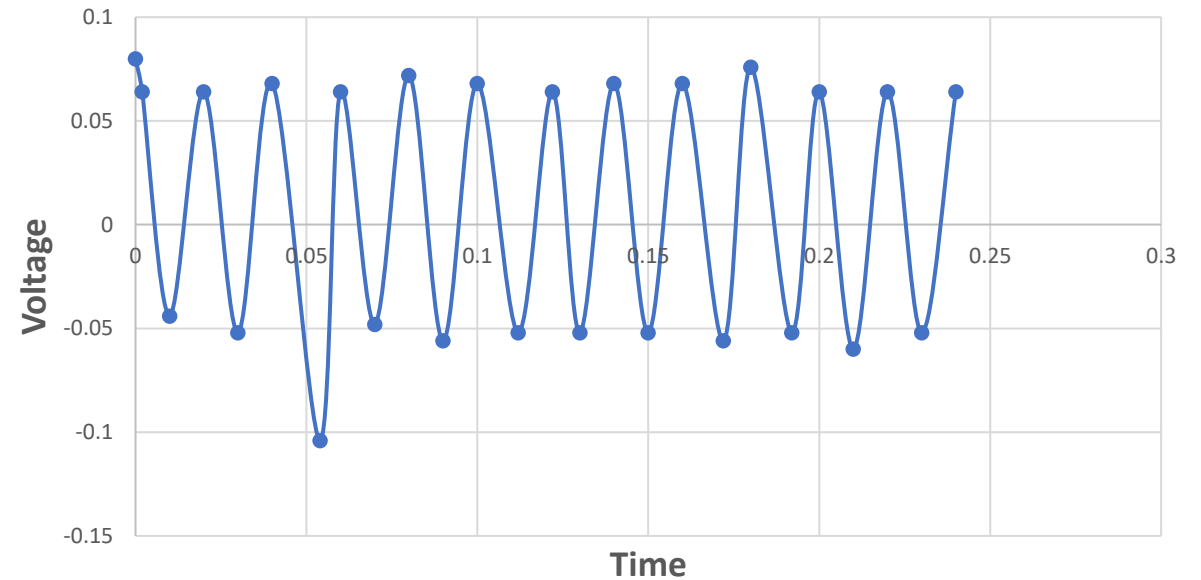
Side-Bottom Result



- Resulting graphs of the side bottom with 100Hz and 10KHz.



Using 100Hz Frequency



Using 10KHz Frequency



Conclusion

- The finger-simplified model was created in model geometry. A framework made up of five parts was used to build the finger. (i.e., skin, fat, artery, muscle, and bone)
- The study includes COMSOL simulation results compared with experimental results to obtain the optimum parameters for this technique. The COMSOL simulation uses 4 electrodes in different positions with a current of 10mA concerning frequency 100Hz to 100GHz.
- It was shown that multiple electrodes can be useful in obtaining impedance change.
- Useful if signal can not be obtained from one pair, or to further support the results.



Conclusion

- The study was performed under normal resting conditions where an experimental procedure was set up to measure the impedance response from the finger.
- Experiment results were not conclusive as more sensitive impedance analyzers are required.
- This research will be helpful to find out the heartbeat using impedance plethysmography.

References of Literature Review

- Morales, I., González-Landaeta, R., & Simini, F. (2021). Bioimpedance plethysmography with capacitive electrodes and sole force sensors: Comparative trial. *Journal of Physics: Conference Series*, 2008(1). <https://doi.org/10.1088/1742-6596/2008/1/012018>
- Piuze, E., Pisa, S., Pittella, E., Podestà, L., & Sangiovanni, S. (2020). Wearable belt with built-in textile electrodes for cardio—respiratory monitoring. *Sensors (Switzerland)*, 20(16), 1–15. <https://doi.org/10.3390/s20164500>
- Ramkumar, M., & Babu, C. G. (2017). A Comprehensive Analysis of Forearm Impedance Plethysmography for the Maximally Non Invasive Monitoring and Measurement of Heart Pumping Function. *SSRN Electronic Journal*, 1(2), 178–184. <https://doi.org/10.2139/ssrn.2942416>



Thank you